

复变综合训练选择填空解析

by 王景永

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试题一

一、

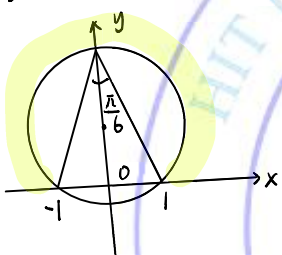
$$1. \quad z = \frac{(1-i)^5 - 1}{(1+i)^3 + 1} = \frac{(1-i)^4(1-i) - 1}{zi(1+i) + 1} = \frac{-4(1-i) - 1}{-1 + 2i} = \frac{(-5 + 4i)(-1 - 2i)}{5}$$

$$= \frac{13 + 6i}{5}$$

$$2. \quad re^{i(\theta + \frac{\pi}{3})} = \sqrt{2} e^{i\frac{\pi}{4}} \quad z = \sqrt{2} e^{-\frac{\pi}{2}} = \sqrt{2} \left(\frac{\sqrt{6} + \sqrt{2}}{4} \right) + i \sqrt{2} \left(\frac{\sqrt{2} - \sqrt{6}}{4} \right) = \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right) + i \left(\frac{1}{2} - \frac{\sqrt{3}}{2} \right)$$

$$3. \quad \arg z = \arg(-1 + \sqrt{3}i) = \arctan(-\sqrt{3}) + \pi = \frac{2}{3}\pi$$

$$4. \quad \sin \frac{\pi}{3} - i \cos \frac{\pi}{3} = \cos(-\frac{\pi}{6}) + i \sin(-\frac{\pi}{6})$$

$$5. \quad \begin{cases} y > 0 \\ x^2 + (y - \sqrt{3})^2 > 4 \end{cases}$$


二、

$$1. \quad C \quad z^2 = \frac{1}{2}(-2i) = -i \quad z^{100} + z^{50} + 1 = -1 - i + 1 = -i$$

2. **B** 有界, $C \times$ 无边界, 不是闭区域, $D \times$, 显然为多连通区域

3. **A**

$$4. \quad C \quad \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x}{1+x^2+y^2} = 0 \quad 0 \leq \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2}{\sqrt{x^2+y^2}} \leq \lim_{x \rightarrow 0} \frac{x^2}{|x|} = 0$$

$$5. \quad D \quad 0 \leq \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{(x^2 - y^2)^2}{x^2 + y^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{(x^2 + y^2)^2}{x^2 + y^2} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^2 + y^2) = 0$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 - y^2}{x^2 + y^2} \begin{cases} x=0 \rightarrow \lim_{y \rightarrow 0} -\frac{y^2}{y^2} = -1 \\ y=0 \rightarrow \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1 \end{cases} \quad \therefore \text{极限不存在}$$

试题二

一.

1. u, v 在 z 平面处处可微

$$\begin{cases} \frac{\partial u}{\partial x} = e^x \cos y - 1 = \frac{\partial v}{\partial y} = e^x \cos y - 2y \\ \frac{\partial u}{\partial y} = -e^x \sin y = -\frac{\partial v}{\partial x} = -e^x \sin y \end{cases} \quad \text{在 } y = \frac{1}{2} \text{ 上可导, 在 } C \text{ 上无处解析}$$

$$\begin{aligned} 2. \quad f(u, v) &= \int_{(0,0)}^{(u,v)} e^u \sin v \, du + e^u \cos v \, dv \\ &= \int_0^v e^u \cos v \, dv = e^u \sin v + C \end{aligned}$$

$$\begin{aligned} 3. \quad \frac{\partial u}{\partial x} &= 2nxy = \frac{\partial v}{\partial y} = 2lx y \\ \frac{\partial u}{\partial y} &= 3my^2 + nx^2 = -\frac{\partial v}{\partial x} = -ly^2 - 3x^2 \end{aligned} \quad \begin{cases} n=l \\ 3m=-l \\ n=-3 \end{cases} \quad \therefore \begin{cases} m=1 \\ n=-3 \\ l=-3 \end{cases}$$

$$4. \quad |\sqrt{3}+i| = e^{(\sqrt{3}+i)2k\pi i} = e^{-2k\pi} \cdot e^{i(2\sqrt{3}k\pi)} = e^{-2k\pi} [\cos(2\sqrt{3}k\pi) + i \sin(2\sqrt{3}k\pi)], k \in \mathbb{Z}$$

$$5. \quad f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

二.

1. D

2. C

$$f(z) = |z|^2 = x^2 + y^2 \quad \lim_{z \rightarrow 0^+} f(z) = \lim_{z \rightarrow 0^-} f(z) = 0, \text{ 连续}$$

$$\begin{cases} \frac{\partial u}{\partial x} = 2x \Big|_{x=0} = \frac{\partial v}{\partial y} = 2y \Big|_{y=0} = 0 \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0 \end{cases} \quad \text{满足 C-R 条件}$$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{x^2 + 2x\Delta x + \Delta x^2 + y^2 + 2y\Delta y + \Delta y^2 - x^2 - y^2}{\Delta x + i\Delta y}$$

$$0 \leq \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \left| \frac{\Delta x^2 + \Delta y^2}{\Delta x + i\Delta y} \right| = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \sqrt{\Delta x^2 + \Delta y^2} = 0 \quad \text{可导}$$

$\forall (0, \delta)$ 内任意一点均不满足 C-R 条件, 均不可导, $f(z)$ 在 $z=0$ 不解析.

3. D

4. D

5. B

$f(z) = u + iv$ 解析, v 是 u 的共轭调和函数, $-u$ 是 v 的共轭调和函数

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试题三

1. $x=t, y=t, 0 \leq t \leq 1 \quad z(t) = (1+i)t \quad z'(t) = 1+i$

$$\int_0^{1+i} (8z^2 + 3\bar{z} + 1) dz = \int_0^1 [8(1+i)^2 t^2 + 3(1-i)t + 1] (1+i) dt = \int_0^1 [16it^2 + 3(1-i)t + 1] (1+i) dt$$

$$= \left\{ \left[\frac{16}{3}it^3 + \frac{3}{2}(1-i)t^2 + t \right] (1+i) \right\}_0^1 = \left(\frac{16}{3}i + \frac{3}{2} - \frac{3}{2}i + 1 \right) (1+i) = -\frac{4}{3} + \frac{19}{3}i$$

$x=t, y=t^2, 0 \leq t \leq 1 \quad z(t) = t+it^2 \quad z'(t) = 1+2ti$

$$\int_0^{1+i} (8z^2 + 3\bar{z} + 1) dz = \int_0^1 [8t^2(1+it)^2 + 3(t-it^2) + 1] (1+2ti) dt$$

$$= \int_0^1 [-16it^5 - 40t^4 + 132it^3 + 61t^3 + 8+3i)t^2 + (2i+3)t + 1] dt = -\frac{8}{3}i - 8 + 8i + \frac{3}{2} + \frac{8}{3} + i + i + \frac{3}{2} + 1 = -\frac{4}{3} + \frac{22}{3}i$$

2. $\oint_{|z|=2} \frac{|z|^2}{z-1} dz = \oint_{|z|=2} \frac{4}{z-1} dz = 4 \times 2\pi i = 8\pi i$

$\oint_{|z|=3} \frac{|z|^2}{z-3} dz = \oint_{|z|=2} \frac{4}{z-3} dz = 0$

3. $f(z) = \oint_{|z|=2} \frac{z(e^z + \sin z)}{z-z} dz$

$|z| < 2 \quad f(z) = 2\pi i z (e^z + \sin z) \Big|_{z=z} = 2\pi i z (e^z + \sin z) \quad f'(z) = 2\pi i [e^z + \sin z + z e^z + z \cos z]$

$f(0) = 0 \quad f'(0) = 2\pi i$

$|z| > 2 \quad f(z) = 0, \quad f'(z) = 0 \quad f'(3) = 0$

4. 由 Cauchy 积分定理 $\oint_{\Gamma} \frac{f'(z) + 2f(z) + 3z}{f^2(z)} dz = 0$

5. $\oint_{|z|=2} \frac{\sin z}{(z-\frac{\pi}{2})^3} dz = \frac{2\pi i}{2!} (\sin z)'' \Big|_{z=\frac{\pi}{2}} = -\pi i$

二.

1. C 2. D 3. C $\oint_{\Gamma} f(z) dz = \oint_{\Gamma} u dx - v dy + i \oint_{\Gamma} v dx + u dy$ (课本 P69)

$\operatorname{Re} \left(\oint_{\Gamma} f(z) dz \right) = \oint_{\Gamma} (\operatorname{Re} f(z)) dx - (\operatorname{Im} f(z)) dy$

4. D A. 奇点在曲线上, 积分不确定 X B. $\oint_{\Gamma} \frac{dz}{z-2} dz = -2\pi i$ X

C. $\oint_{\Gamma} \frac{dz}{z-2} dz = 2\pi i$ X D. $\left\{ \begin{array}{l} \text{均在外, 由 Cauchy 积分定理} \oint_{\Gamma} \frac{dz}{(z-1)(z-2)} dz = 0 \\ \text{均在内部} \oint_{\Gamma} \frac{dz}{(z-1)(z-2)} dz = -2\pi i + 2\pi i = 0 \end{array} \right.$ ✓

5. B

试题四

一、

1. $\lim_{n \rightarrow \infty} a_n = a_0$ 且 $\lim_{n \rightarrow \infty} b_n = b_0$

2. 引理: 幂级数的收敛半径是其和函数的所有奇点到原点的最短距离. (辅导 P137)

$\frac{e^{\frac{z}{z-2}} \sin z}{(z^2+3z+2)e^z}$ 奇点有 $z=-2, z=-1, z=i \therefore R=1$

$\sum_{n=0}^{\infty} a_n z^n$ 在 $|z| < 1$ 绝对收敛, 在 $|z| \leq r, (0 < r < 1)$ 一致收敛

3. $R=3$

4. $\lim_{n \rightarrow \infty} \left| \frac{C_{n+1}}{C_n} \right| = \frac{1}{R} \quad \lim_{n \rightarrow \infty} \left| \frac{\frac{C_{n+1}}{b^{n+1}}}{\frac{C_n}{b^n}} \right| = \left| \frac{C_{n+1} b^n}{C_n b^{n+1}} \right| = \left| \frac{1}{Rb} \right| \therefore R' = R|b|$

5. $\begin{cases} 0 < \frac{1}{|z-2|} < 1 \\ 0 < \left| 1 - \frac{z}{2} \right| < 1 \end{cases} \therefore 1 < |z-2| < 2$

$$\sum_{n=1}^{\infty} \frac{1}{(z-2)^n} + \sum_{n=0}^{\infty} (-1)^n \left(1 - \frac{z}{2} \right)^n = \frac{1}{1 - \frac{1}{z-2}} - 1 + \frac{1}{1 + 1 - \frac{z}{2}}$$

$$= \frac{1}{z-3} + \frac{2}{4-z}$$

二

1. D 2. B 3. C 4. B proof: $\sum_{n=0}^{\infty} C_n z^n$ 在 $z=1$ 收敛, $\therefore \sum_{n=0}^{\infty} C_n z^n$ 在 $|z| < 1$ 内

绝对收敛 $\therefore R \geq 1$, 假设 $R > 1$, 则 $\sum_{n=0}^{\infty} C_n z^n$ 在 $z=1$ 绝对收敛, 即 $\sum_{n=0}^{\infty} |C_n|$ 收敛, 矛盾.

故 $R=1$

5. C 课本 P84, 高阶导数公式要求 D 为单联通域, 题中为多联通域, 公式不成立

试题五

一.

$$1. \lim_{z \rightarrow \pi} \frac{\sin z}{z - \pi} = \lim_{z \rightarrow \pi} \frac{\cos z}{1} = -1 \quad \text{可去奇点}$$

2. 6阶极点

3. 可去奇点

$$4. \operatorname{Res} \left[\frac{e^z - 1}{z^5}, 0 \right] = \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d^3}{dz^3} \left(\frac{e^z - 1}{z} \right) = \frac{1}{3!} \lim_{z \rightarrow 0} \frac{d^3}{dz^3} \left(1 + \frac{z}{2!} + \frac{z^2}{3!} + \frac{z^3}{4!} + \dots \right) \Big|_{z=0} = \frac{1}{24}$$

$$5. \operatorname{Res} \left[\frac{z^2 + 1}{e^z}, \infty \right] = -\operatorname{Res} \left[\frac{z^2 + 1}{z^4 e^{\frac{1}{z}}}, 0 \right] = 0$$

$$\frac{z^2 + 1}{z^4} \cdot e^{-\frac{1}{z}} = \left(\frac{1}{z^4} + \frac{1}{z^2} \right) \left(1 - \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \dots \right) \quad \therefore a_{-1} = 0$$

$$6. \operatorname{Res} [\cot^2 z, 0] = \lim_{z \rightarrow k\pi} \frac{d}{dz} \frac{(z - k\pi)^2 \cos^2 z}{\sin^2 z} = \lim_{z \rightarrow k\pi} \left[\frac{(z - k\pi)^2}{\sin^2(z - k\pi)} \right]'$$

$$= \lim_{z \rightarrow 0} \left(\frac{z^2}{\sin^2 z} \right)' = \lim_{z \rightarrow 0} \frac{2z \sin^2 z - 2 \sin z \cos z \cdot z^2}{\sin^4 z} = 2 \lim_{z \rightarrow 0} \frac{1 - \cos z}{z} = 0$$

二.

1. B 2. B $f(z) = \frac{z^7}{(z-1)(1-z^2)^2} = \frac{z^7}{z^2(1-z)(z^2-1)^2} \quad \therefore \infty \text{ 为二阶极点}$

3. D $f(z) = \frac{1}{\cos \frac{1}{z-1}} = \frac{1}{\cos \xi}$

$z \rightarrow 1, \xi \rightarrow \infty$, 此时找不到一个 $\dot{U}(\xi, \delta)$ 是处处解析的

$\therefore z=1$ 为非孤立奇点.

4. C 答案错误

5. D $0, \frac{3}{2}, \pm \frac{\pi}{2}$

6. A $\operatorname{Res} \left[\frac{z}{z^4 - 1}, \infty \right] = -\operatorname{Res} \left[\frac{z}{1 - z^4}, 0 \right]$

$$\frac{z}{1 - z^4} = z (1 + z^4 + z^8 + \dots) \quad \therefore a_{-1} = 0$$

一.

$$1. \quad z = 1 + it, \quad -1 \leq t \leq 1 \quad w = z^3 = -3t^2 + i(3t - t^3) \quad \begin{cases} u = 1 - 3t^2 \\ v = t(3 - t^2) \end{cases} \quad -1 \leq t \leq 1$$

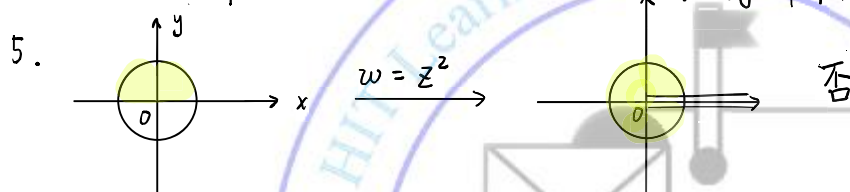
$$2. \quad |f'(-1+2i)| = |2z+4|_{z=-1+2i} = 2\sqrt{5} \quad \theta = \arg f'(-1+2i) = \arg(2+4i)$$

3. 半径为有限值的圆周 直线

4. 如果 $w = \frac{\alpha z + \beta}{\gamma z + \delta}$ 将 $|z|=1$ 映为直线, 那么直线上的无穷远点的原像必在单位圆上,

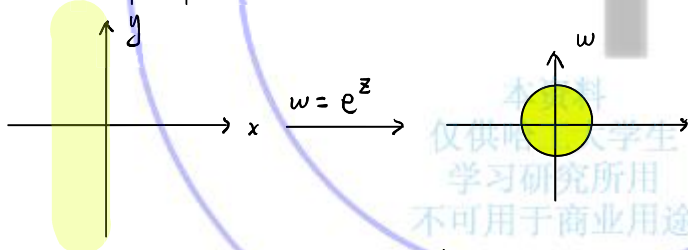
$$\text{则 } z|_{w=\infty} = \frac{\delta w - \beta}{-\gamma w + \alpha}|_{w=\infty} = -\frac{\delta}{\gamma} \text{ 必在单位圆上 } (\Leftrightarrow) \quad \left| \frac{\delta}{\gamma} \right| = 1 \text{ 或 } |\delta| = |\gamma|$$

$$\text{又 } z' = \frac{\alpha \delta - \beta \gamma}{(\alpha - \gamma w)^2} \neq 0 \quad \therefore \alpha \delta - \beta \gamma \neq 0 \quad \therefore \begin{cases} |\delta| = |\gamma| \\ \alpha \delta - \beta \gamma \neq 0 \end{cases}$$



二

1. **B** $|w'| = |e^z| < 1, \quad \operatorname{Re} z < 0$



虚部影响辐角
实部影响半径

2. **C** 保角映射: $\begin{cases} w = f(z) \text{ 在区域 } D \text{ 解析} \\ z_0 \text{ 为 } D \text{ 内一点, } f'(z_0) \neq 0 \end{cases}$

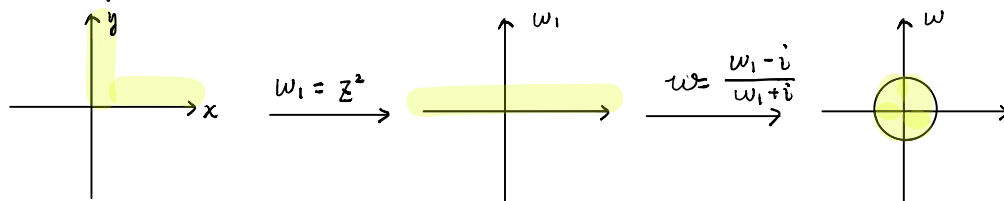
A. 不解析 $\times$ B. 不解析 $\times$ D. $w' = \frac{1}{z^2}, \quad z \rightarrow \infty, \quad w' \rightarrow 0 \times$

3. **A** $-1 \rightarrow \frac{1}{2} \quad i \rightarrow \frac{1}{2} - \frac{1}{2}i \quad -i \rightarrow \frac{1}{2} + \frac{1}{2}i, \quad z \rightarrow 1, \quad z \text{ 被映为 } -\infty \quad \therefore \operatorname{Re} z < \frac{1}{2}$

4. **D** 设对称点 $z = x + iy$

$$2 \cdot \sqrt{x^2 + (y-1)^2} = 9, \quad B, C \text{ 排除, } A \text{ 与 } 2+i \text{ 在圆心两侧, 排除}$$

5. **C**



试题七

一.

$$1. \sin^3 t = \frac{3}{4} \sin t - \frac{1}{4} \sin 3t$$

$$F[\sin^3 t] = \frac{3}{4} i\pi [\delta(\omega+1) - \delta(\omega-1)] - \frac{1}{4} \pi i [\delta(\omega+3) - \delta(\omega-3)]$$

$$2. F[f(at-t_0)] = F[f(a(t-\frac{t_0}{a}))] = e^{-\frac{i\omega t_0}{a}} \frac{1}{|a|} F(\frac{\omega}{a})$$

$$3. F[u(t-\tau)] = e^{-i\omega\tau} F[u(t)] = e^{-i\omega\tau} [\frac{1}{i\omega} + \pi\delta(\omega)]$$

$$4. F[e^{-\beta|t|}] (\beta > 0) = \frac{2\beta}{\beta^2 + \omega^2} \quad F^{-1}[\frac{2}{1+\omega^2}] = \frac{1}{2} e^{-|t|}$$

$$5. \int_{-\infty}^{\infty} \cos \omega t dt = \int_{-\infty}^{\infty} \frac{e^{i\omega t} + e^{-i\omega t}}{2} dt = \frac{1}{2} [2\pi\delta(\omega) + 2\pi\delta(-\omega)] = 2\pi\delta(\omega)$$

$$6. \int_{-4}^2 e^t \delta(t+3) dt = \int_{-\infty}^{+\infty} e^{t-3} \delta(t) dt = e^{-3}$$

二

$$1. A \quad 2. C \quad 3. B \quad F[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} e^{-\beta|t|} e^{-i\omega t} dt$$

$$= \int_{-\infty}^0 e^{(\beta-i\omega)t} dt + \int_0^{\infty} e^{-(\beta+i\omega)t} dt = \frac{1}{\beta-i\omega} + \frac{1}{\beta+i\omega} = \frac{2\beta}{\beta^2 + \omega^2}$$

$$4. A \quad F[f(t) \sin \omega_0 t] = \int_{-\infty}^{\infty} f(t) \sin \omega_0 t e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(t) \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i} e^{-i\omega t} dt$$

$$= \frac{1}{2i} \left(\int_{-\infty}^{\infty} f(t) e^{-i(\omega-\omega_0)t} dt - \int_{-\infty}^{\infty} f(t) e^{-i(\omega+\omega_0)t} dt \right)$$

$$= \frac{1}{2i} [F(\omega-\omega_0) - F(\omega+\omega_0)] = \frac{i}{2} [F(\omega+\omega_0) - F(\omega-\omega_0)]$$

$$5. D \quad F[u(t) \sin \omega_0 t] = \frac{i}{2} \left[\frac{1}{i(\omega+\omega_0)} - \frac{1}{i(\omega-\omega_0)} + \pi\delta(\omega+\omega_0) - \pi\delta(\omega-\omega_0) \right]$$

$$= \frac{\omega_0}{\omega_0^2 - \omega^2} + \frac{\pi i}{2} [\delta(\omega+\omega_0) - \delta(\omega-\omega_0)]$$

$$6. B \quad \int_{-\infty}^{\infty} \delta(t-3) (t^2+1) dt \xrightarrow{u=t-3} \int_{-\infty}^{\infty} \delta(u) (u^2+6u+10) du$$

$$= (u^2+6u+10)|_{u=0} = 10$$

试题八

- 一.
- $L[f(at-b)u(at-b)] = L[f(a(t-\frac{b}{a}))u(a(t-\frac{b}{a}))] = e^{-s\frac{b}{a}}\frac{1}{a}F(\frac{s}{a})$
 - $L[t^2\cos 2t] = \frac{d^2}{ds^2} L[\cos 2t] = \frac{d^2}{ds^2} (\frac{s}{s^2+4}) = \frac{2s^3-24s}{(s^2+4)^3}, \text{Res} > 0$
 - $L[\frac{\sin 3t}{t}] = \int_0^{+\infty} L[\sin 3t] ds = \int_s^{+\infty} \frac{3}{u^2+9} du = \frac{\pi}{2} - \arctan(\frac{s}{3})$
 - $L^{-1}[\arctan \frac{1}{s}] = L^{-1}[\int_s^{+\infty} \frac{-1}{1+u^2} du] = \frac{L^{-1}[\frac{1}{1+s^2}]}{t} = \frac{\sin t}{t}$
 - $L^{-1}[\ln(1+\frac{1}{s^2})] = -2L^{-1}[\int_s^{+\infty} \frac{1}{u} - \frac{u}{u^2+1} du] = 2\frac{L^{-1}[\frac{1}{s}] - L^{-1}[\frac{s}{s^2+1}]}{t} = \frac{2(1-\cos t)}{t}$
 - $L^{-1}[\frac{1+e^{-2s}}{s^2}] = L^{-1}[\frac{1}{s^2}] + L^{-1}[\frac{e^{-2s}}{s^2}] = t + (t-2)u(t-2)$

- 二.
- C
 - B
 - A $L[\sin(t-\frac{\pi}{3})] = \frac{1}{2}L[\sin t] - \frac{\sqrt{3}}{2}L[\cos t] = \frac{1-\sqrt{3}s}{2(1+s^2)}$

question: 为什么不能使用延时性质?

$$L[\sin(t-\frac{\pi}{3})] = e^{-\frac{\pi}{3}s} L[\sin t] = \frac{e^{-\frac{\pi}{3}s}}{1+s^2} ? \text{ 原因见后论文}$$

$$4. B \quad L[\int_0^t \tau e^{a\tau} \sin a\tau d\tau] = \frac{L[t e^{at} \sin at]}{s} = \frac{-\frac{d}{ds}[L(e^{at} \sin at)]}{s}$$

$$= -\frac{\frac{d}{ds}[\frac{a}{(s-a)^2+a^2}]}{s} = \frac{2(s-a)a}{s[a^2+(s-a)^2]^2}$$

$$5. D \quad 6. A \quad L^{-1}[sF'(s)] = \frac{d}{dt}[L^{-1}(F'(s))] = \frac{d}{dt}[-t L^{-1}(F(s))]$$

$$= \frac{d}{dt}[-tf(t)] = -f(t) - tf'(t)$$

综合一

一.

$$1. f(x, y) = (x + iy)y + x = x(y+1) + iy^2 \quad u = x(y+1) \quad v = y^2$$

显然 u, v 在 C 上处处连续, $\therefore f(z)$ 在 C 上处处连续

$$\begin{cases} \frac{\partial u}{\partial x} = y+1 = \frac{\partial v}{\partial y} = 2y \\ \frac{\partial u}{\partial y} = x = -\frac{\partial v}{\partial x} = 0 \end{cases} \quad \therefore \begin{cases} y=1 \\ x=0 \end{cases} \quad \therefore \text{处处不解析}$$

$$2. (1+i)^i = [\sqrt{2} e^{(\frac{\pi}{4} + 2k\pi)i}]^i = e^{-(\frac{\pi}{4} + 2k\pi) + \frac{i}{2} \ln 2} = e^{-(\frac{\pi}{4} + 2k\pi)} [\cos \frac{\ln 2}{2} + i \sin \frac{\ln 2}{2}]$$

$$3. \text{当 } |z| < 2 \text{ 时, } f(z) = \oint_{|\zeta|=2} \frac{\sin(\frac{\pi}{3}\zeta)}{\zeta - z} d\zeta = 2\pi i \sin(\frac{\pi}{3}z) \Big|_{\zeta=z} = 2\pi i \sin(\frac{\pi}{3}z)$$

$$f'(z) = \frac{2\pi^2 i}{3} \cos(\frac{\pi}{3}z) \quad \therefore f'(1) = \frac{\pi^2 i}{3}$$

$$4. f(z) = (z-1)^2 e^{\frac{1}{z}} = (z^2 - 2z + 1) \left(1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots \right) = \left(\frac{1}{3!} - \frac{2}{2!} + 1 \right) \frac{1}{z} + \dots = \frac{1}{6z} + \dots$$

$$\text{Res}[f(z), 0] = \frac{1}{6}$$

$$5. F[f(3t-2)] = F[f(3(t-\frac{2}{3}))] = \frac{1}{3} e^{-\frac{2}{3}\omega i} F\left(\frac{\omega}{3}\right)$$

二.

$$1. C \quad \arg z = \arctan \frac{\sin \alpha}{1 - \cos \alpha} = \arctan \frac{1}{\tan \frac{\alpha}{2}} = \frac{\pi - \alpha}{2}$$

$$2. C \quad 3. B \quad \begin{cases} 0 < \frac{2}{|z-3|} < 1 \\ 2 < |z-3| < 3 \\ 0 < |1 - \frac{z}{3}| < 1 \end{cases}$$

$$4. D \quad f(z) = (z-1)^2 \sin \frac{1}{z-1} = (z-1)^2 \left(\frac{1}{z-1} - \frac{1}{3!(z-1)^3} + \dots \right) = (z-1) - \frac{1}{3!(z-1)} + \dots$$

$\therefore z=1$ 为 $f(z)$ 本性奇点

$$5. C \quad F[e^{i2t} \delta'(t)] = i\omega F[e^{i2t} \delta(t)] = i(\omega - 2)$$

综合三

一.

$$1. re^{i(0+\frac{\pi}{2})} = \sqrt{2} e^{i\frac{\pi}{4}} \quad z = \sqrt{2} e^{-\frac{\pi}{12}} = \sqrt{2} \left(\frac{\sqrt{6}+\sqrt{2}}{4} \right) + i\sqrt{2} \left(\frac{\sqrt{2}-\sqrt{6}}{4} \right) = \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right) + i \left(\frac{1}{2} - \frac{\sqrt{3}}{2} \right)$$

$$2. f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

$$3. f(z) = \oint_{|\zeta|=2} \frac{\zeta(e^{\zeta} + \sin \zeta)}{\zeta - z} d\zeta$$

$$|z| < 2 \quad f(z) = 2\pi i \zeta (e^{\zeta} + \sin \zeta) \Big|_{\zeta=z} = 2\pi i z (e^z + \sin z) \quad f'(z) = 2\pi i [e^z + \sin z + z e^z + z \cos z]$$

$$f(0) = 0 \quad f'(0) = 2\pi i$$

$$4. \begin{cases} 0 < \frac{1}{|z-2i|} < 1 \\ 0 < |1 - \frac{z}{2}| < 1 \end{cases} \quad \therefore 1 < |z-2i| < 2$$

5. 本性奇点

二

$$1. D \quad w' = \frac{4i}{(z+i)^2} \quad w' \Big|_{z=2i} = \frac{4i}{-9} \quad \arg w' \Big|_{z=2i} = -\frac{\pi}{2}$$

$$2. A \quad \text{Res} \left[\frac{z}{z^4-1}, \infty \right] = -\text{Res} \left[\frac{z}{1-z^4}, 0 \right]$$

$$\frac{z}{1-z^4} = z(1+z^4+z^8+\dots) \quad \therefore a_{-1} = 0$$

$$3. C \quad F[(t-2)f(t)] = iF'(\omega) - 2F(\omega)$$

$$4. B \quad \text{设对称点 } z = x+iy$$

$$2 \cdot \sqrt{x^2 + (y-1)^2} = 9$$

$$5. A$$

综合四

一、

$$1. i^{\sqrt{2}} = e^{[i(\frac{\pi}{2} + 2k\pi)]\sqrt{2}} = \cos[\sqrt{2}(\frac{\pi}{2} + 2k\pi)] + i\sin[\sqrt{2}(\frac{\pi}{2} + 2k\pi)], k=0, \pm 1, \pm 2 \dots$$

$$\ln(-1-i) = \ln\sqrt{2} + i\pi(\frac{-3}{4})$$

$$\sin(2+3i) = \operatorname{ch}3 \cdot \sin 2 + i\operatorname{sh}3 \cdot \cos 2 \quad \text{这个题... 没有什么意义...}$$

$$2. 2\pi i \quad \text{见综合(三) } - - - 3$$

$$3. \sum_{n=0}^{\infty} (-1)^n (n+1) (z-1)^n, |z-1| < 1$$

4. 收敛

$$\begin{aligned} 5. F\left[\int_{-\infty}^t u(t) e^{-t} \sin^2 t dt\right] &= \frac{F[u(t)] e^{-t} \sin^2 t}{i\omega} + \pi F(0) \delta(\omega) \\ &= \frac{F[u(t)] \cdot e^{-t} (1 - \cos 2t)}{2i\omega} + \pi F(0) \delta(\omega) \\ &= \frac{F[u(t)] e^{-t} (1 - \frac{1}{2} e^{-2ti} - \frac{1}{2} e^{2ti})}{2i\omega} + \pi F(0) \delta(\omega) \\ &= \frac{1}{2i\omega} \left\{ \frac{1}{i(\omega-i)} + \pi \delta(\omega-i) - \frac{1}{2i(\omega+2-i)} - \frac{1}{2} \pi \delta(\omega+2-i) - \frac{1}{2i(\omega-2-i)} \right. \\ &\quad \left. - \frac{1}{2} \pi \delta(\omega-2-i) \right\} + \pi F(0) \delta(\omega) \\ &= \frac{1}{2i\omega} \left\{ \frac{1}{i\omega+1} - \frac{i(\omega-i)}{4-(\omega-i)^2} + \pi \delta(\omega-i) - \frac{1}{2} \pi [\delta(\omega+2-i) + \delta(\omega-2-i)] \right\} + \pi \left[\frac{2}{5} \right] \delta(\omega) \end{aligned}$$

答案错误

$$\begin{aligned} * F(0) &= \lim_{t \rightarrow +\infty} \left(\int_{-\infty}^t u(t) e^{-t} \sin^2 t dt \right) = \lim_{t \rightarrow +\infty} \left(\int_0^t e^{-t} \sin^2 t dt \right) \\ &= \left[-\frac{1}{2} e^{-x} - \frac{e^{-x}}{2} \left(\frac{2 \sin 2x - \cos 2x}{5} \right) \right] \Big|_0^{+\infty} = \frac{2}{5} \end{aligned}$$

二、

$$1. D \quad 2. C \quad C \text{项要求 } D \text{ 为单联通域} \quad 3. A$$

$$4. B \quad \text{课本 } P_{84}, \text{ 高阶导数公式要求 } D \text{ 为单联通域, 题中为多联通域, 公式不成立}$$

$$\begin{aligned} 5. A \quad L^{-1}[sF(s)] &= \frac{d}{dt} [L^{-1}(F(s))] = \frac{d}{dt} [-t L^{-1}(F(s))] \\ &= \frac{d}{dt} [-t f(t)] = -f(t) - t f'(t) \end{aligned}$$

综合五

1. u, v 在 z 平面处处可微

$$\begin{cases} \frac{\partial u}{\partial x} = e^x \cos y - 2x = \frac{\partial v}{\partial y} = e^x \cos y - 2 \\ \frac{\partial u}{\partial y} = -e^x \sin y = -\frac{\partial v}{\partial x} = -e^x \sin y \end{cases} \quad \text{在 } x=1 \text{ 上可导, 在 } C \text{ 上无处解析}$$

2. 引理: 幂级数的收敛半径是其和函数的所有奇点到原点的最短距离. (辅导 P137)

$$\frac{e^{\frac{z}{z-2}} \sin z}{(z^2+3z+2)e^z} \quad \text{奇点有 } z=-2, z=-1, z=i \quad \therefore R=1$$

$\sum_{n=0}^{\infty} a_n z^n$ 在 $|z| < 1$ 绝对收敛, 在 $|z| > 1$ 发散

$$3. \begin{cases} 0 < \frac{1}{|z-2|} < 1 \\ 0 < |1 - \frac{z}{2}| < 1 \end{cases} \quad \therefore 1 < |z-2| < 2$$

$$\sum_{n=1}^{\infty} \frac{1}{(z-2)^n} + \sum_{n=0}^{\infty} (-1)^n \left(1 - \frac{z}{2}\right)^n = \frac{1}{1 - \frac{1}{z-2}} - 1 + \frac{1}{1 + 1 - \frac{z}{2}} = \frac{1}{z-3} + \frac{2}{4-z}$$

$$4. \quad L[t^2 \cos 2t] = \frac{d^2}{ds^2} L[\cos 2t] = \frac{d^2}{ds^2} \left(\frac{s}{s^2+4} \right) = \frac{2s^3 - 24s}{(s^2+4)^3}, \quad \text{Res} > 0$$

$$\begin{aligned} 5. \quad F(\omega) &= F[f(t)] = \int_{-\infty}^{+\infty} e^{-i\omega t} \cos t e^{-i\omega t} dt = \int_{-\infty}^{+\infty} e^{-i\omega t} \frac{e^{it} + e^{-it}}{2} e^{i\omega t} dt \\ &= \frac{1}{2} \left\{ \int_0^{+\infty} e^{[-1+i(1-\omega)]t} dt + \int_{-\infty}^0 e^{[1+i(1-\omega)]t} dt + \int_0^{+\infty} e^{[-1-i(1+\omega)]t} dt + \int_{-\infty}^0 e^{[1-i(1+\omega)]t} dt \right\} \\ &= \frac{1}{2} \left[\frac{1}{1-i(1-\omega)} + \frac{1}{1+i(1-\omega)} + \frac{1}{1-i(1+\omega)} + \frac{1}{1+i(1+\omega)} \right] = \frac{2\omega^2+4}{\omega^2+4} \end{aligned}$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{2\omega^2+4}{\omega^2+4} e^{i\omega t} d\omega = \frac{1}{\pi} \int_0^{+\infty} \frac{2\omega^2+4}{\omega^2+4} \cos \omega t d\omega$$

$$\therefore \int_0^{+\infty} \frac{\omega^2+2}{\omega^2+4} \cos \omega t d\omega = \frac{\pi}{2} f(t) = \frac{\pi}{2} e^{-|t|} \cos t$$

1. A 2. D 3. B 课本 P84, 高阶导数公式要求 D 为单联通域 4. D

$$5. C \quad f(z) = \frac{z^7}{(z-1)(1-z^2)^2} \xrightarrow{z=\frac{1}{z}} \frac{1}{z^2(1-\frac{1}{z})(\frac{1}{z^2}-1)^2} \quad \therefore \infty \text{ 为二阶极点}$$

综合六

一.

$$1. (1+i)^{\sqrt{2}} = e^{\sqrt{2} \ln(1+i)} = e^{\sqrt{2} [\ln\sqrt{2} + i\frac{\pi}{4} + i2k\pi]} = \sqrt{2}^{\sqrt{2}} e^{\sqrt{2}i(\frac{\pi}{4} + 2k\pi)}, k \in \mathbb{Z}$$

$$\ln(2+i) = \ln\sqrt{5} + i\arctan\frac{1}{2} \quad \sin(1+2i) = \sin 1 \cosh 2 + i \sinh 2 \cdot \cos 2$$

$$2. C \text{ 内连续 } y = x^2 + 1 \quad C \text{ 内处处不解析}$$

$$3. \infty \quad a_{-m} \quad \text{可去奇点}$$

$$4. F[e^{-\beta|t|}] (\beta > 0) = \frac{2\beta}{\beta^2 + \omega^2} \quad F^{-1}\left[\frac{3}{4 + \omega^2}\right] = \frac{3}{4} e^{-|t|}$$

$$5. f(z) = \oint_{|\zeta|=2} \frac{\zeta(e^{\zeta} + \cos \zeta)}{\zeta - z} d\zeta$$

$$|z| < 2 \quad f(z) = 2\pi i \zeta (e^{\zeta} + \cos \zeta) \Big|_{\zeta=z} = 2\pi i z (e^z + \cos z) \quad f'(z) = 2\pi i [e^z + \cos z + z e^z - z \sin z]$$

$$f(0) = 0 \quad f'(0) = 4\pi i$$

$$|z| > 2 \quad f(z) = 0 \quad \therefore f'(4) = 0$$

二.

$$1. A \quad 2. B \quad 3. C$$

$$4. A \quad B, C \text{ 不一定}, D: f(z) = u + iv \text{ 解析, } v \text{ 是 } u \text{ 的共轭调和函数, } -u \text{ 是 } v \text{ 的共轭调和函数}$$

$$5. C \quad F[(t+3)f(t)] = iF'(\omega) + 3F(\omega)$$